

Pharmaceutical Case Study

AsterNova Pharmaceuticals: Balancing Growth, Safety, Adherence and Quality

Background Story

AsterNova Pharmaceuticals is a fictional mid-sized pharmaceutical company operating across India and selected international markets. In 2025, the company launched Cardiomab-X, a fictional once-daily oral medicine intended for adults with inadequately controlled hypertension despite standard first-line therapy. The product had completed a staged development programme and entered the market after regulatory review. The case below is fictional, but its operating pressures are grounded in real pharmaceutical concerns: clinical evidence, patient safety, medication adherence, manufacturing quality, pharmacovigilance, access, and post-market monitoring.

The Situation

During the first six months after launch, AsterNova saw rapid physician adoption in urban hospitals. Sales exceeded forecast, but the pharmacovigilance and medical-affairs teams noticed three emerging issues. First, treatment discontinuation was higher in patients reporting dizziness. Second, refill data suggested that adherence fell over time. Third, one manufacturing site reported an increase in packaging-line deviations, although released batches continued to meet final quality specifications. Senior management had to decide whether to prioritize commercial expansion, adherence support, additional safety studies, or manufacturing investment.

Real-World Grounding

The scenario reflects broader realities of medicine use and development. The World Health Organization has reported that adherence to long-term therapy for chronic disease is often a major challenge, historically estimating average adherence around 50% in developed countries. WHO also identifies medication-related harm as a major component of preventable harm in health care. Drug development normally progresses from discovery and preclinical research through clinical research, regulatory review, and post-market safety monitoring. These facts provide context for the fictional company data below.

Fictional Case Dataset

Metric	Pre-launch / Trial	Month 3	Month 6
Patients exposed	1,200	8,500	22,000
Mean systolic BP reduction	13.8 mmHg	13.1 mmHg	12.7 mmHg
Patients reaching target BP	68%	65%	63%
Reported dizziness	6.2%	7.1%	7.8%
Serious adverse-event reports	0.8%	0.9%	1.0%
Estimated medication adherence	82%	74%	66%
Treatment discontinuation	9%	12%	16%
On-time monthly refills	—	76%	68%
Packaging-line deviations / month	2	4	7
Units sold (cumulative)	—	310,000	790,000